

Airbus Delays Development of Hydrogen-Powered Aircraft

European manufacturer Airbus has announced a delay in its development efforts of a hydrogen-powered aircraft. The company explained that progress on the project has been slower and more complex than expected.

Airbus had set a goal to make a zero-emission airplane by 2035. The company has not announced a new target date for the plane's production. But the French union Force Ouvrière said some of its aircraft workers were recently told the technology was running five to 10 years behind, Reuters news agency reported.

The French news agency AFP reported Airbus also said it had reduced the program's budget by 25 percent, but was still examining ways to reduce emissions.

"Our ambition and roadmap towards decarbonizing. . . remain unchanged," an Airbus spokesman told AFP.

In September 2020, Airbus presented three hydrogen-fueled airplane designs, called ZEROe, as part of its efforts to decarbonize.

A model of the Airbus Zero-e aircraft is pictured at the ILA Berlin Air Show 2022 in Berlin, Germany June 22, 2022.

Other airplane manufacturers, however, have centered more on renewable fuels or developing electric-powered planes for short-distance flights.

Airbus noted it has no plans to completely stop developing hydrogen-powered aircraft technology. The company said it sees hydrogen as "a revolutionary energy source for aviation."

However, Airbus said in a statement to Reuters it recognizes that developing systems to produce, transport, and store hydrogen represent "a huge challenge requiring global collaboration and investment."

Difficulties with hydrogen

Hydrogen motors do not release gases that can warm the earth's atmosphere. Instead, they release water.

However, much of the hydrogen used for fuel is produced using natural gas or coal. Both of these production methods often release carbon dioxide gas, which can warm the atmosphere.

FILE -- P V R Murthy, center, general manager at Oil India Limited, pump station 3, shows a part of a hydrogen plant in Jorhat, India, Thursday, Aug. 17, 2023.

Running renewable electricity through water can produce hydrogen without adding to carbon emissions. However, this process is generally more costly. Transporting and storing hydrogen also present problems.

As a gas at room temperature, hydrogen takes up a very large space. This makes it impossible to use as fuel. It first needs to be turned into a liquid to be transported or stored. And it needs to be cooled to -253° Celsius to be liquefied. This cooling process requires its own source of energy. Liquefied hydrogen still takes up four times the space of jet fuel.

The airline industry trade association IATA believes aviation will reach most of its reduction in emissions by using sustainable aviation fuel (SAF). It believes other technologies, like hydrogen, will take longer to develop.

SAF uses food and plant waste from cities, agricultural and forest wastes, and other non-petroleum sources. This fuel is then added to jet fuel in mixtures of 10 to 50 percent to become SAF.

Airbus says it agrees that SAF can also play an important part in reducing emissions, particularly for medium and long-distance flights.

I'm Andrew Smith.

Andrew Smith adapted this story for VOA Learning English based on reporting from Agence France-Presse and Reuters.

Vocabulary

emission -n. การปล่อยก๊าซ

ambition -n. ความทะเยอทะยาน, ความมุ่งมั่นที่จะบรรลุเป้าหมาย

roadmap -n. แผนปฏิบัติการเพื่อบรรลุเป้าหมาย

renewable -adj. ซึ่งหมุนเวียนได้, ซึ่งนำกลับมาใช้ใหม่ได้ (เช่น ทรัพยากรลม น้ำ แสงอาทิตย์ ที่มีอยู่ตลอดเวลา)

challenge -n. ความท้าทาย, สิ่งที่ยากจะทำหรือเป้าหมายที่ยากจะไปถึง

collaboration -n. การทำงานร่วมกัน, การร่วมมือกัน

sustainable -adj. ยั่งยืน, ซึ่งสามารถใช้ซ้ำได้โดยไม่ก่อให้เกิดมลพิษเพิ่มขึ้น

petroleum -n. น้ำมันปิโตรเลียม, น้ำมันดิบ (ที่พบใต้พื้นผิวโลกหรือทะเล)

Reading Comprehension Quiz

1. What is the main announcement made by European manufacturer Airbus?

They have successfully developed a hydrogen-powered aircraft.

They are launching a new zero-emission aircraft by 2030.

They are delaying the development of their hydrogen-powered aircraft.

They are shifting their focus to electric-powered planes.

2. What was Airbus's original target year for creating a zero-emission airplane?

2025

2030

2035

2040

3. According to the French union Force Ouvrière, how far behind is the hydrogen technology running?

One to two years

Three to five years

Five to ten years

More than ten years

4. How much has Airbus reduced the program's budget for emission reduction efforts?

10 percent

15 percent

25 percent

50 percent

5. What did an Airbus spokesman state about the company's ambition and roadmap towards decarbonizing?

They are being re-evaluated due to budget cuts.

They remain unchanged despite the delays.

They have been accelerated due to new technologies.

They are now focused primarily on sustainable aviation fuel.

6. What was the name given to the three hydrogen-fueled airplane designs presented by Airbus in September 2020?

HydroGenX

EcoFlyer

ZEROe

CleanSky

7. What is the primary focus of *other* airplane manufacturers mentioned in the article?

Developing hydrogen-powered aircraft.

Centering on renewable fuels or electric-powered planes for short-distance flights.

Improving traditional jet fuel efficiency.

Collaborating with Airbus on hydrogen technology.

8. How does Airbus view hydrogen as an energy source for aviation?

As a temporary solution until better technologies emerge.

As a “revolutionary energy source.”

As a source with too many insurmountable difficulties.

As a viable option only for short-distance flights.

9. According to Airbus, what major challenge does developing hydrogen systems represent?

The cost of raw materials.

A lack of skilled engineers.

A “huge challenge requiring global collaboration and investment.”

Public skepticism about its safety.

10. What do hydrogen motors release instead of gases that warm the earth's atmosphere?

Oxygen

Nitrogen

Water

Carbon dioxide

11. What is a common problem associated with the *production* of much of the hydrogen currently used for fuel?

It requires extremely rare minerals.

It often releases carbon dioxide gas.

It is too noisy for urban areas.

It demands excessive amounts of water.

12. What is the main drawback of producing hydrogen by running renewable electricity through water?

It is generally more costly.

It requires specialized, expensive equipment.

It produces harmful byproducts.

It is only possible in certain geographical locations.

13. What is required to make hydrogen suitable for transport or storage as fuel?

It needs to be heated to a high temperature.

It must be combined with natural gas.

It needs to be turned into a liquid by cooling it to -253° Celsius.

It can only be transported through specialized pipelines.

14. According to the airline industry trade association IATA, which technology is expected to achieve most of the aviation industry's emission reduction?

Hydrogen-powered aircraft.

Electric-powered planes.

Sustainable aviation fuel (SAF).

Advanced aerodynamics.

15. From what sources does Sustainable Aviation Fuel (SAF) primarily obtain its components?

Directly from petroleum-based sources.

Food and plant waste, agricultural/forest wastes, and other non-petroleum sources.

Specially cultivated algae and synthetic compounds.

Recycled engine lubricants and industrial solvents.